



Figure 1: Main screen of MOVE

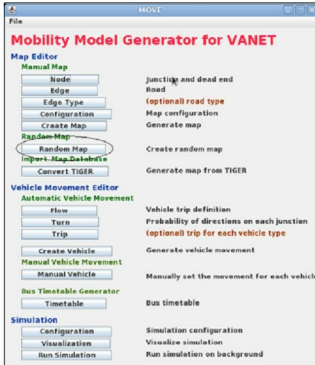


Figure 2: Screen for mobility modelling

simulation into an NS2-readable form, after which further experiments can be conducted for network traffic analysis.

To generate the SUMO simulation, click 'Mobility Model'. Figure 2 shows the available mobility generation functionality. As an example of these, Figure 3 shows the creation of the node database. Unlike XML files, only the table shown in the figure is required to be furnished; no XML tags need to be scripted. Similarly, data for other files is furnished in tables (see Figure 3) by selecting the relevant file from the screen shown in Figure 2. Again, if there is a need to generate road traffic of hundreds of nodes, instead of filling the tables, there is an auto-generation functionality (highlighted by a circle in Figure 2).

Once the simulation is configured, it can be viewed by using the visualisation tab. SUMO can be then used to generate communication traffic simulation on NS2. The traffic model tab in Figure 1 will be used to generate the NS2 simulation. Figure 4 shows the window where NS2-readable files will be configured. You need to load the SUMO trace file created after the previously stated process. This will yield the screen shown in Figure 5. In the table on the right side of the screen, network traffic can be specified.

Thus, using MOVE, you can easily generate a SUMO simulation and convert it to NS2-readable form. For more



Figure 3: Table for inserting nodes in SUMO simulation

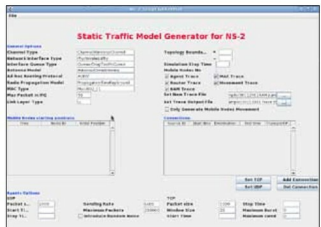


Figure 4: Screen for network traffic simulation generation

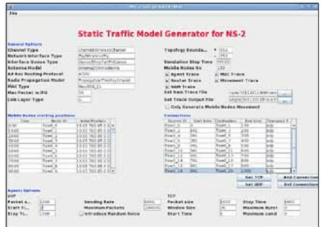


Figure 5: Screen after loading the SUMO files

details, visit <http://www.cs.unsw.edu.au/klan/move/> for MOVE and <http://sumo.sourceforge.net/> for SUMO. 

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